

Mapping Self-Illumination

Self-Illumination maps create the illusion of incandescence by replacing any shadows on the surface with the diffuse colour. This is good for simulating glowing surfaces and can also be used to remove shadowed areas on an object.

Mapping Bumps

Bump maps give a surface a sense of roughness and depth. Most surfaces are not perfectly smooth, and a little bit of bump mapping can go a long way toward making objects seem real. It is also easier and more efficient to add surface imperfections by using Bump or Displacement maps than to actually model every bump and pockmark on a surface.

Bump mapping only affects the way light interacts with the surface, but does not actually modify geometry; so the edges still appear smooth. Bump mapping is perfectly fine for creating general surface roughness or more-exaggerated bumps on surfaces such as a ground plane or a wall. Bump mapping affects the character of a surface but does not modify the geometry.

Mapping Displacement

Displacement mapping changes the surface of the geometry, adding another level of realism. This process takes more processing and rendering power, so use displacements only where they are needed and use bumps for the rest. Because displacement mapping physically changes the geometry, the spinner to the left of the map is the height of the displacement. Displacement also distorts the actual geometry of the object, so higher-resolution geometry might be needed.

Mapping Reflections

Any shiny object reflects light. Reflections are important in simulating reality. A mirror might reflect an image perfectly, whereas galvanized metal might reflect only broad swaths of colour, but they both reflect at least a little bit of their environment. The most accurate form of reflections comes from using the Raytrace map, but many artists simply use bitmapped environments. Although not as accurate, bitmapped environments can be much faster to render, and, in most cases, reflections do not need to be completely accurate to sell the illusion of reality. If there is no “real” environment, such as when an object is rendered by itself, bitmap environments are a necessity. They are particularly useful when creating animation that integrates with film or video. Artists who work on special effects often take photographs of the film sets in order to generate bitmap reflections.